

MODULE DESCRIPTION

A. PHASE 1

i. User Module

Manages all users in the system such as:

- Allowing users to register using details (name, phone number and password).
- Allowing users to log in and log out

ii. Trade Module

This is responsible for creating and managing transactions between a buyer and a seller.

It allows the buyer to initiate a trade by entering details such as the item, amount and seller information. The seller then reviews and accepts or rejects the request.

Once the seller accepts, the trade becomes active and the system records the agreement between both parties.

This module ensures that both buyer and seller agree on the terms before any payment is made and it acts as the starting point for the escrow process.

iii. Escrow Module

Holds the buyer's money securely during a transaction.

When the buyer makes payment, the money is locked in the system and not given to the seller immediately. The funds are only released to the seller after the agreed conditions are met, such as delivery confirmation.

If the seller fails to deliver, the money is refunded to the buyer automatically.

iv. Smart Contract Module (Ethereum)

Automatically controlling how money is handled in the system.

It contains rules written in code (on Ethereum) that decide when to release or refund money.

These rules are executed automatically without human involvement.

B. PHASE 2

v. Transaction Module

Responsible for recording and tracking all transactions in the system.

It stores details of every trade and shows the status of each transaction, such as pending, completed and failed.

Users can view their transaction history and monitor progress at any time.

vi. Dispute Module

Responsible for handling disagreements between the buyer and seller.

It allows users to raise a complaint, submit evidence and have the issue reviewed before a final decision is made.

Based on the decision, the system will either release the money to the seller or refund the buyer

vii. Notification Module

Responsible for keeping users updated about their transactions. It sends alerts when important events happen, such as:

- Payment received
- Trade accepted
- Money released
- Dispute updates

viii. Rating & Review Module

Stores feedback and ratings. This module also helps build trust between users by allowing users to rate each other after a trade.